EXPERIMENT 7

CALCULATION AND VERIFICATION OF FREQUENCY DEVIATION AND MODULATION INDEX OF FM SIGNAL

AIM

To calculate and verify the frequency deviation and modulation index of a Frequency Modulation (FM) signal using MATLAB.

SOFTWARE REQUIRED

MATLAB

PROGRAM CODE

```
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signal
clc;
clear;
close all;

%% Message Signal Parameters
Am = 2;       % Message amplitude (V)
fm = 500;     % Message frequency (Hz)

%% Carrier Signal Parameters
Ac = 1;       % Carrier amplitude (V)
fc = 10000;   % Carrier frequency (Hz)

%% FM Parameters
kf = 250;     % Frequency sensitivity (Hz/V)
fs = 100000;  % Sampling frequency (Hz)
T = 0.01;     % Simulation duration (s)

%% Time Vector
t = 0:1/fs:T;

%% Message Signal
m = Am*cos(2*pi*fm*t);

%% Frequency Deviation
delta_f = kf*Am;

%% Modulation Index
beta = delta_f/fm;

%% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

%% Display Parameters
fprintf('Carrier Frequency (fc) = %.0f Hz\n',fc);
fprintf('Message Frequency (fm) = %.0f Hz\n',fm);
fprintf('Message Amplitude (Am) = %.2f V\n',Am);
fprintf('Frequency Sensitivity (kf) = %.2f Hz/V\n',kf);
fprintf('Frequency Deviation (Delta f) = %.2f Hz\n',delta_f);
fprintf('Modulation Index (beta) = %.2f\n',beta);

%% Verification
beta_verify = delta_f/fm;
fprintf('Verified Modulation Index = %.2f\n',beta_verify);

%% Verification Check
tolerance = 1e-6;
if abs(beta - beta_verify) < tolerance
    fprintf('Verification Successful\n');
else
    fprintf('Verification Failed\n');
end

%% Plot Message and FM Signals
figure;
subplot(2,1,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
grid on;
```

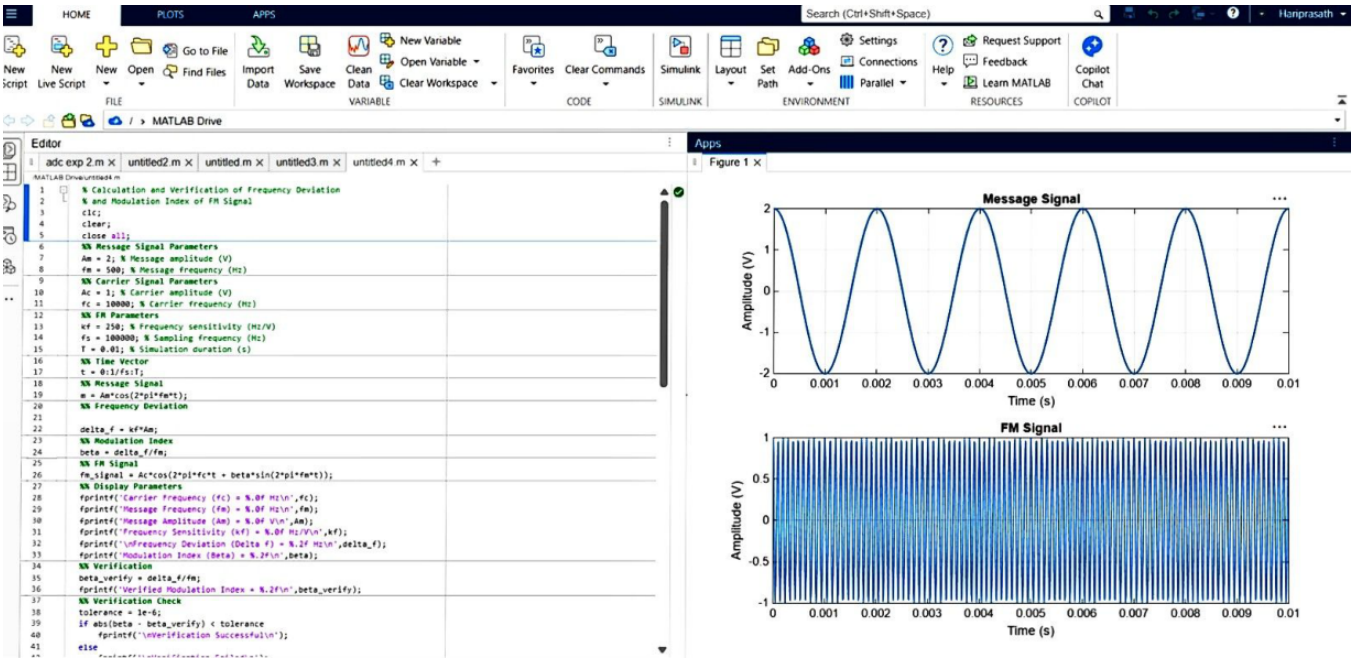
```
subplot(2,1,2);
plot(t,fm_signal);
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
grid on;
```

PROCEDURE

1. Open MATLAB and create a new script.
2. Enter the program code and run the script.
3. Observe the calculated frequency deviation and modulation index in the Command Window.
4. Check the verification message and observe the message and FM waveforms.

OUTPUT

Original MATLAB screenshot supplied by the student:



For the given values, frequency deviation = kf × Am = 250 × 2 = 500 Hz, and modulation index β = Δf/fm = 500/500 = 1.

RESULT

The frequency deviation and modulation index of the FM signal were calculated and verified using MATLAB.